

Miscoding the shared component of a bridged network costs identifiability, not accuracy; miscoding an internal component biases the bridged estimate

A simulation study answering catalog problem CMP-06

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Abstract

Background. Component network meta-analysis parses treatment labels into components deterministically (1), and a disconnected network is bridged when its subnetworks share a component. Catalog problem CMP-06 asks what a coding error does there: perturb an estimate, or change what is estimable.

Methods. Two subnetworks linked only by component A; the target is the effect of a component observed only across the gap. Codings in the second subnetwork were wrong independently with probability up to 0.3, either coding the shared A as a distinct agent (connectivity-changing) or dropping an internal component (connectivity-preserving). Deterministic fits with a rank screen, coding sensitivity and probabilistic coding; 2000 replicates per cell.

Results. Coding the shared component as distinct removed the bridge's information from that trial: the target became non-estimable in 9.5% of replicates at the highest rate, which the rank screen detected exactly, and otherwise stayed unbiased with coverage 0.943 given a miscoding. Dropping an internal component biased the target by -0.105 and cut coverage to 0.826 given a miscoding, with the row space unchanged. Probabilistic coding restored coverage (0.967 or more); coding sensitivity did too but returned the whole line whenever a plausible coding broke the bridge.

Conclusion. For a real bridge, the dangerous miscoding is the one that leaves the network's shape intact. A rank screen over the plausible codings is free and catches the other kind; propagating coding uncertainty handles the biasing kind.

1 The problem

A target contrast is estimable if it lies in the row space of the design implied by the component coding. Coding a shared component as two distinct agents can remove the only path across the gap; omitting a component from an arm leaves the row space unchanged but attributes that component's effect to the others. The two errors are different in kind, and which one a plausible miscoding produces is computable before fitting.

2 Design

Registered protocol: `protocol.md`. Subnetwork 1: X vs X+A (two trials), X vs X+C. Subnetwork 2: Y+A vs Y+B+D (two trials), Y vs Y+D. Component effects A -0.3, B -0.2, C -0.1, D -0.15; each contrast has variance 0.01. The target, the effect of B, is identified under the true coding only through A. Each subnetwork-2 trial is miscoded with probability p : A coded as a distinct A2, or D dropped from the Y+B+D

arm. Coding sensitivity takes the union of the intervals over every plausible coding of those two trials, or the whole line if one of them leaves B unidentified. Probabilistic coding mixes the plausible codings' estimates by their prior probability (at least 0.05 per trial).

3 Results

Table 1: 2000 replicates per cell; deterministic coverage is conditional on the target being estimable under the coding used. Coverage MCSE at most 0.008 over all replicates; up to 0.025 in the given-a-miscoding column.

miscod- ing	p	target non- estimable	determinis- tic coverage	given a miscoding	bias given a miscoding	sensitivity coverage (bounded)	probabilis- tic coverage (bounded)
shared A coded as distinct	0.00	0.0%	0.951			1.000 (0.0%)	0.952 (100.0%)
shared A coded as distinct	0.05	0.4%	0.952	0.945	-0.021	1.000 (0.0%)	0.957 (100.0%)
internal D dropped	0.05	0.0%	0.944	0.868	-0.095	0.980 (100.0%)	0.967 (100.0%)
shared A coded as distinct	0.15	2.3%	0.951	0.944	0.001	1.000 (0.0%)	0.965 (100.0%)
internal D dropped	0.15	0.0%	0.923	0.840	-0.103	0.979 (100.0%)	0.969 (100.0%)
shared A coded as distinct	0.30	9.5%	0.952	0.943	0.004	1.000 (0.0%)	1.000 (0.0%)
internal D dropped	0.30	0.0%	0.885	0.826	-0.105	0.983 (100.0%)	0.967 (100.0%)

The registered primary was refuted for inference (Table 1): when the bridge is real, coding the shared component as distinct in one trial costs that trial's bridging information but not accuracy, and coding it so in both makes the target non-estimable, which the rank screen reports without fitting. The second null control failed, and that is the finding: the miscoding that leaves the row space intact was the one that biased the estimate, by about 0.8 of its standard error, because the omitted component's effect was attributed to B. Both uncertainty methods covered it. Coding sensitivity paid for this with uninformative output in every connectivity-changing cell, since its plausible set always included a coding without the bridge; probabilistic coding stayed bounded unless that coding carried more than 5% prior probability.

A false bridge, a distinct agent coded as the shared component, is the reverse error. The target is then not identified under the true coding, so every fitted interval is a model extrapolation whose error is the effect difference between the two agents; this was not simulated, and no coding method fitted to the data can detect it.

4 What this does not answer

One hand-built network, contrast-level fixed-effect fits, no covariate adjustment, synergy or Bayesian arm, independent miscoding within a two-trial plausible set. Peer review has not been done.

References

1. Gerta Rücker, Maria Petropoulou, Guido Schwarzer. Network meta-analysis of multicomponent interventions. *Biometrical Journal*. 2020;62(3):808–21. doi:[10.1002/bimj.201800167](https://doi.org/10.1002/bimj.201800167)